Motilal Nehru National Institute of Technology Allahabad
Prayagraj-211004 [India]

Computer Science and Engineering Department
Mid Semester (Odd) Examination 2026-27 (MOCK)

Programme Name: B.Tech
Course Code: CSN15403

Semester: V
Course Name: Cryptography and Network Security
Student Reg. No.: _____

Duration: 90 Minutes

Max. Marks: 25

Instructions: (Related to Questions)

1. Precise answer with proper justification will be considered to get complete marks.
2. Attempt all questions.

		Marks
Q1	**a.** In each of the following, what is the maximum number of characters that will be changed in the ciphertext if only a single character is changed in the plaintext? (i) Hill cipher with a 3×3 key (ii) Keyed transposition cipher (iii) Auto-key cipher (iv) DES in ECB mode, 8-character (64-bit) blocks. **b.** A hash function produces a 64-bit digest. How many digests must Eve create to find a collision with probability more than 0.5? Compare with a preimage attack.	2
Q2	Eve secretly gets access to Alice's computer and, using her cipher, types `hi`. The screen shows `CL`. Eve knows that Alice is using an affine cipher. a. What type of attack is Eve launching? b. Find the key pair (k_1, k_2). c. Decrypt the intercepted ciphertext `VBBGRGENNE`.	3
Q3	Use the Playfair cipher to encipher the message "*Hide the keys in the classroom*". The key matrix is made from the keyword "*MOTILAL NEHRU*" followed by the rest of the alphabet. Assume 'X' as the bogus letter and that I/J go in the same cell.	4
Q4	In GF(2^8), find the inverse of $(x^7 + 1)$ modulo $(x^8 + x^4 + x^3 + x + 1)$ using the extended Euclidean algorithm. Also state: 1- How many iterations your method takes. 2- Value of the remainder in iteration 2. 3- Value of the quotient in iteration 3.	4
Q5	Solve the following: i. In the field GF(13), find the result of $7 \div 5$. ii. Find 3^{2025} mod 13 and 7^{-1} mod 40 (show the extended Euclidean table). iii. Find the particular and general solutions of the linear Diophantine equation $12x+18y = 30$.	3
Q6	**a.** Bob's RSA public key is $(e, n) = (5, 323)$. Eve intercepts the ciphertext $C = 264$. Showing every step, find Bob's private key d and the plaintext M. Why would this attack fail for a 2048-bit n? **b.** For the elliptic curve $E_{11}(1, 6) : y^2 = x^3 + x + 6 \pmod{11}$, compute $2P$ for $P = (2, 7)$ and verify that the result lies on the curve.	5

Q7 In the Advanced Encryption Standard, the state after SubBytes in round 1 and the round-1 key are: 4

$$S = \begin{bmatrix} \texttt{d4} & \texttt{e0} & \texttt{b8} & \texttt{1e} \\ \texttt{27} & \texttt{bf} & \texttt{b4} & \texttt{41} \\ \texttt{11} & \texttt{98} & \texttt{5d} & \texttt{52} \\ \texttt{ae} & \texttt{f1} & \texttt{e5} & \texttt{30} \end{bmatrix} \qquad K_1 = \begin{bmatrix} \texttt{a0} & \texttt{88} & \texttt{23} & \texttt{2a} \\ \texttt{fa} & \texttt{54} & \texttt{a3} & \texttt{6c} \\ \texttt{fe} & \texttt{2c} & \texttt{39} & \texttt{76} \\ \texttt{17} & \texttt{b1} & \texttt{39} & \texttt{05} \end{bmatrix}$$

a. Show the ShiftRows transformation.
b. Compute the first column of the MixColumns output (show the GF(2^8) working).
c. Given that MixColumns produces the state below, perform AddRoundKey.

$$\begin{bmatrix} \texttt{04} & \texttt{e0} & \texttt{48} & \texttt{28} \\ \texttt{66} & \texttt{cb} & \texttt{f8} & \texttt{06} \\ \texttt{81} & \texttt{19} & \texttt{d3} & \texttt{26} \\ \texttt{e5} & \texttt{9a} & \texttt{7a} & \texttt{4c} \end{bmatrix}$$

Model Answers

Every number below was computed and checked by script; Q7 values are the FIPS-197 Appendix B test vectors.

Q1 (2 marks)

a. (i) **3**: the whole 3-letter block depends on every plaintext letter in it. (ii) **1**: transposition only moves characters, so only the changed one changes (it just sits at another position). (iii) **2**: C_i changes, and P_i is the key for C_{i+1} ($k_{i+1} = P_i$), so C_{i+1} changes too. (iv) **8**: the avalanche effect spreads one changed bit across the whole 64-bit block.
b. Collision = birthday problem 3: $k = 1.18 \times 2^{n/2} = 1.18 \times 2^{32} \approx 5.07 \times 10^{9}$ digests. Preimage = problem 1: $k = 0.69 \times 2^{64} \approx 1.27 \times 10^{19}$. A 64-bit digest is safe against preimage but **not** against collision.

Q2 (3 marks)

a. Chosen-plaintext attack (Eve chooses the plaintext and sees the ciphertext).
b. $h = 7 \to C = 2$, $i = 8 \to L = 11$: $7k_1 + k_2 \equiv 2$, $8k_1 + k_2 \equiv 11 \pmod{26}$. Subtract: $k_1 \equiv 9$; then $k_2 \equiv 2 - 63 = -61 \equiv 17$. $\gcd(9, 26) = 1$, so $(k_1, k_2) = (\mathbf{9}, \mathbf{17})$ is valid.
c. $P = (C - 17) \times 9^{-1}$, $9^{-1} = 3$ ($27 \equiv 1$): V= 21: $(21 - 17) \cdot 3 = 12 =$ m; B= 1: $(1 - 17) \cdot 3 = -48 \equiv 4 =$ e; ...
VBBGRGENNE → **meetatnoon** ("meet at noon").

Q3 (4 marks)

Key matrix (MOTILAL NEHRU → M O T I L A N E H R U, then the rest):

M	O	T	I/J	L
A	N	E	H	R
U	B	C	D	F
G	K	P	Q	S
V	W	X	Y	Z

Digrams: hi de th ek ey si nt he cl as sr **ox** om (oo in "classroom" is split with the filler X; 26 letters so no final filler).

hi	de	th	ek	ey	si	nt	he	cl	as	sr	ox	om
col	rect	rect	rect	rect	rect	rect	row	rect	rect	col	rect	row
DH	**CH**	**IE**	**NP**	**HX**	**QL**	**EO**	**RH**	**FT**	**RG**	**ZF**	**TW**	**TO**

Ciphertext: **DHCHIENPHXQLEORHFTRGZFTWTO**.
Checks: hi (same column, each moves down: h→D, i→H); he (same row, each moves right: h→R, e→H); om (same row: o→T, m→O); sr (same column S/R: s→Z, r→F).

Q4 (4 marks)

$r_1 = x^8 + x^4 + x^3 + x + 1$, $r_2 = x^7 + 1$, $t_1 = 0$, $t_2 = 1$; each row $t = t_1 + q\,t_2$ (subtraction = addition in GF(2)).

It.	q	r_1	r_2	r	t_1, t_2	t
1	x	$x^8 + x^4 + x^3 + x + 1$	$x^7 + 1$	$x^4 + x^3 + 1$	0, 1	x
2	$x^3 + x^2 + x + 1$	$x^7 + 1$	$x^4 + x^3 + 1$	$x^2 + x$	1, x	$x^4 + x^3 + x^2 + x + 1$
3	x^2	$x^4 + x^3 + 1$	$x^2 + x$	1	x, $t^{(2)}$	$x^6 + x^5 + x^4 + x^3 + x^2 + x$
4	$x^2 + x$	$x^2 + x$	1	0		

($t^{(2)}$ = the t of iteration 2.) Final $r_1 = 1 \Rightarrow$ inverse exists: $(x^7 + 1)^{-1} = \mathbf{x^6 + x^5 + x^4 + x^3 + x^2 + x}$ (7E). Check: 81 ⊗ 7E = 01 ✓.

1- **4** iterations. 2- Remainder in iteration 2: $\mathbf{x^2 + x}$. 3- Quotient in iteration 3: $\mathbf{x^2}$.

Q5 (3 marks)

i. $7 \div 5 = 7 \times 5^{-1}$; $5 \times 8 = 40 \equiv 1 \pmod{13}$, so $5^{-1} = 8$; $7 \times 8 = 56 \equiv \mathbf{4}$.

ii. Fermat: $3^{12} \equiv 1 \pmod{13}$, $2025 = 168 \cdot 12 + 9$, so $3^{2025} \equiv 3^9 = (3^3)^3 = 27^3 \equiv 1^3 = \mathbf{1}$.

$7^{-1} \bmod 40$:

q	r_1	r_2	r	t_1	t_2	t
5	40	7	5	0	1	-5
1	7	5	2	1	-5	6
2	5	2	1	-5	6	-17
2	2	1	0	6	-17	40
	1	0		-17		

$\Rightarrow 7^{-1} \equiv -17 \equiv \mathbf{23}$ ($7 \cdot 23 = 161 = 4 \cdot 40 + 1$).

iii. $d = \gcd(12, 18) = 6 \mid 30$: infinitely many solutions. Divide: $2x + 3y = 5$; $2(-1) + 3(1) = 1$, so $s = -1$, $t = 1$. Particular: $x_0 = 5(-1) = -5$, $y_0 = 5(1) = 5$. General: $\mathbf{x = -5 + 3k}$, $\mathbf{y = 5 - 2k}$ ($k \in \mathbb{Z}$).
Check: $12(-5) + 18(5) = 30$ ✓.

Q6 (5 marks)

a. Factor n: trial division by primes $\leq \sqrt{323} \approx 17.97$ gives $323 = \mathbf{17} \times \mathbf{19}$. $\phi(n) = 16 \times 18 = 288$.
$d = 5^{-1} \bmod 288$: rows $(q; r_1, r_2, r; t)$: $(57; 288, 5, 3; -57)$, $(1; 5, 3, 2; 58)$, $(1; 3, 2, 1; -115)$, $(2; 2, 1, 0)$
$\Rightarrow d \equiv -115 \equiv \mathbf{173}$ ($5 \cdot 173 = 865 = 3 \cdot 288 + 1$).
$M = 264^{173} \bmod 323 = \mathbf{42}$. Check by re-encrypting: $42^2 = 1764 \equiv 149$, $42^4 \equiv 149^2 \equiv 237$, $42^5 \equiv 237 \cdot 42 = 9954 \equiv 264 \pmod{323}$ ✓.
For a 2048-bit n (two ≈1024-bit primes), factoring is computationally infeasible, so $\phi(n)$ and d cannot be found: RSA's security is exactly the factoring problem.

b. Doubling: $s = \dfrac{3x_1^2 + a}{2y_1} = \dfrac{3 \cdot 4 + 1}{14} \equiv 13 \cdot 3^{-1} \equiv 2 \cdot 4 = 8 \pmod{11}$ (since $14 \equiv 3$, $3^{-1} = 4$).
$x_3 = s^2 - 2x_1 = 64 - 4 = 60 \equiv 5$; $y_3 = s(x_1 - x_3) - y_1 = 8(2 - 5) - 7 = -31 \equiv 2$. So $\mathbf{2P} = (\mathbf{5}, \mathbf{2})$.
Check: $y^2 = 4$; $x^3 + x + 6 = 125 + 5 + 6 = 136 \equiv 4 \pmod{11}$ ✓.

Q7 (4 marks)

a. Row r rotates left by r bytes:

$$\begin{bmatrix} \texttt{d4} & \texttt{e0} & \texttt{b8} & \texttt{1e} \\ \texttt{bf} & \texttt{b4} & \texttt{41} & \texttt{27} \\ \texttt{5d} & \texttt{52} & \texttt{11} & \texttt{98} \\ \texttt{30} & \texttt{ae} & \texttt{f1} & \texttt{e5} \end{bmatrix}$$

b. Column $(\texttt{d4}, \texttt{bf}, \texttt{5d}, \texttt{30})$; $02 \cdot b$ = shift left, XOR $\texttt{1b}$ if the top bit was 1; $03 \cdot b = 02 \cdot b \oplus b$.
$02 \cdot \texttt{d4} = \texttt{b3}$, $03 \cdot \texttt{bf} = \texttt{da}$: $\texttt{b3} \oplus \texttt{da} \oplus \texttt{5d} \oplus \texttt{30} = \mathbf{04}$
$\texttt{d4} \oplus 02 \cdot \texttt{bf}(65) \oplus 03 \cdot \texttt{5d}(\texttt{e7}) \oplus \texttt{30} = \mathbf{66}$
$\texttt{d4} \oplus \texttt{bf} \oplus 02 \cdot \texttt{5d}(\texttt{ba}) \oplus 03 \cdot \texttt{30}(50) = \mathbf{81}$
$03 \cdot \texttt{d4}(67) \oplus \texttt{bf} \oplus \texttt{5d} \oplus 02 \cdot \texttt{30}(\texttt{60}) = \mathbf{e5}$ $\Rightarrow$ first column $(\texttt{04}, 66, 81, \texttt{e5})$ ✓ (matches part c).

c. XOR element by element with K_1:

$$\begin{bmatrix} \texttt{a4} & \texttt{68} & \texttt{6b} & \texttt{02} \\ \texttt{9c} & \texttt{9f} & \texttt{5b} & \texttt{6a} \\ \texttt{7f} & \texttt{35} & \texttt{ea} & \texttt{50} \\ \texttt{f2} & \texttt{2b} & \texttt{43} & \texttt{49} \end{bmatrix}$$

(e.g. $\texttt{04} \oplus \texttt{a0} = \texttt{a4}$, $66 \oplus \texttt{fa} = \texttt{9c}$)

Trap

Q1(iii): Forouzan's auto-key uses the *previous plaintext* letter as the next key, so it is 2, not "all following characters". Q3: don't forget to split oo in "classroom". Q4: in GF(2^n), $t = t_1 - qt_2 = t_1 + qt_2$ (XOR). Q6a: d is the inverse mod $\phi(n) = 288$, not mod 323.